

KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

The v5 reduction — restricted set, freely fitted, Gaia applied afterwards

Prepared 2026-09-09. Solution: `~/KMTdata/Results/v5/IFfinal_<field>.mat` (variable IFsys; the logical Frame in the same file marks the calibration stars).

1. A short description

The analysis is restricted to a clean calibration set — bright, isolated, and astrometrically well behaved in Gaia — plus the target and three comparison stars at the target's own magnitude. Every one of those stars is **freely fitted** and every one takes part in determining the per-epoch frame at every step. Gaia data are used only to *select* the set, and then again at the very end, as a post-hoc tie that fixes the frame's gauge. Nothing is held at a catalogue value during the fit.

2. Object selection, step by step

Both fields, from the raw matched-source file down to what is actually fitted:

step	BLG41	BLG01
raw sources in the MSc file	1177	1187
survive the standard quality cuts	594	621
have an OGLE I magnitude	540	478
in $14 < I < 17$	292	253
... and isolated: no $I < 18$ companion within 2.0 arcsec	144	127
... and matched to a Gaia DR3 source	144	115
... and that source has a RUWE	142	111
... and RUWE < 1.4 → the calibration set	126	96
+ the target + 3 comparison stars	+4	+4
= analysis set actually fitted	130	100

Notes on the individual cuts:

- The **quality cuts** are the pipeline's standard ones and are not part of this design. They remove sources detected in a median of 10% of epochs, against 80% for those retained; every source found in more than half the epochs survives. The restriction imposed by v5 is therefore **130 of 594**, not 130 of 1177.
- The **magnitude window** costs about half the OGLE-matched sources. It is the single largest cut after the standard ones.
- The **isolation cut** costs another half: 292 → 144 on BLG41. The companion list is the OGLE catalogue, which resolves about six times more sources than KMT detects, so it finds close pairs that our own source list has merged.

- The **RUWE cut** removes 11% (BLG41) and 14% (BLG01) of what reaches it, consistent with the 87% pass rate for $G < 17$ across the whole cone. Crowding is not biasing it heavily.
- BLG01 loses 12 sources at the Gaia-match step where BLG41 loses none. This follows its poorer OGLE coverage, 478 of 621 against 540 of 594.
- **The target is not in the calibration set initially** — at $l = 18.13$ it falls outside the magnitude window — **but rather added explicitly after the selection.**

3. Gaia

Gaia enters twice, and in neither case during the fit.

Selecting the set. RUWE is **not** in the local catsHTM DR3 table, which carries `astrometric_chi2_al` and `astrometric_n_good_obs_al` but neither RUWE nor the `u0(G, BP-RP)` table needed to normalise UWE into it. The true values were queried from the Gaia archive — 9464 sources in a 3 arcmin cone, cached in `~/KMTdata/GaiaRef/`. This mattered: at $G < 17$, **87%** pass RUWE < 1.4 , where the unnormalised UWE proxy predicted 74%.

Fixing the gauge, afterwards. A free fit determines relative astrometry only: applying any global affine to all source positions, and compensating in the per-epoch transformations, leaves every residual unchanged. Six parameters are therefore pure gauge, and the same freedom appears one time-derivative up in the proper motions as a constant plus a linear gradient across the field. The tie removes it. The gauge found was:

	$\mu_\alpha \cos \delta$	μ_δ	calibrators
BLG41	-0.219	+4.758	130
BLG01	+2.834	+3.983	100

mas/yr. With it removed, the seven plotted stars agree between the two fields to 0.25 to 0.78 mas/yr in RA and 0.02 to 0.87 in Dec.

A caution on the tie method. `ml.util.gaiaTie` fits a 2 by 2 matrix from a proper-motion regression. On these small sets it is poorly determined: it returned rotations of **146.34 deg (BLG41) and 108.65 (BLG01)**, 38 deg apart, where the direct positional Gaia-to-pixel map puts the two cut-outs 0.126 deg apart. Its linear part is fitting noise. The gauge above was therefore taken the other way round: the linear part from the positional map, which is solid, and only the constant from the proper-motion comparison. The two methods differ by 0.2 to 0.6 mas/yr on the target, which is a systematic worth carrying.

4. The chain, and every effect fitted in it

The two steps. Both act on the 130 / 100 source subset:

1. **Full-decade SysRem correction** — 2 components on the astrometric residuals, computed once over the whole run and saved. 2. **Final solution**, reusing that correction.

The joint proper-motion step of the reference reduction is not used: with so few, so well-measured sources the motions are solved directly in the global fit. Neither is any per-season fitting — seasons enter only in reporting.

	BLG41	BLG01
step 1	16.3 min	14.8 min
step 2	30.0 min	28.0 min

convergence over the calibration stars, last 3 iterations

0.0041 mas

0.0002 mas

Each step runs two passes internally, and which terms are active differs between them. This matters for reading the saved objects: `IFsys.AnnualEffect` and `IFsys.PixPhase` both read 0 in the v5 solutions, which does **not** mean those effects were ignored — they were fitted and subtracted in pass 1 and switched off for pass 2, which works on data from which they have already been removed.

	pass 1	between	pass 2 (final)
per-epoch affine	fitted		fitted
per-source position and motion	fitted		fitted
DCR / chromatic refraction	fitted		fitted again
annual term	fitted and subtracted from the data		forced off
pixel-phase term	fitted and subtracted from the data		forced off
SysRem		applied	
iterative reweighting	15 iterations		6 iterations

`runIterDetrend` sets `AnnualEffect = false` and `PixPhase = false` whenever it is called with `FinalStep`, which `runIterDetrendMSc` passes only on pass 2. The corrected object from pass 1 is what pass 2 receives, so nothing is lost.

Per epoch — the frame. A full 2-D affine, **6 parameters per exposure**, held in `ParE` as `[a11-1, a12, tx, a21, a22-1, ty]`; the linear rows store the deviation from the identity, so all-zero means "already aligned". The six degrees of freedom are translation (2), rotation (1), uniform scale (1) and shear (2). In v5 this block is **6 × 17 332 = 103 992 free parameters on BLG41** and 105 732 on BLG01, against 520 and 400 for every source parameter combined — the frame outnumbers the astrometry two hundred to one. Each epoch's six parameters are solved from 130 sources, so 260 equations for 6 unknowns; the reference reduction has 1188, which is the origin of v5's noisier frame and of the target's 4 to 12% penalty.

Per source — the astrometry we want. Position (x_0 , y_0) at JD0 = 2019-06-01 and proper motion (μ_x , μ_y), **4 parameters per star** in `ParS`, in pixels and pixels per year at 400 mas/pix. In v5 every one of these is free: `ParSFixed` is empty, which is the single difference from v4.

Differential chromatic refraction. The atmosphere refracts blue light more than red, displacing a star along the parallactic angle by an amount that grows with airmass and depends on its colour. Modelled per axis as `[1, sin(pa)·secz, cos(pa)·secz]` plus six higher-order terms in `(secz·sin pa)n` and `(secz·cos pa)n` for $n = 2, 3, 4$ — **9 per axis, 18 in total** — and fitted separately in **6 equal-population colour bins**. All 130 and 100 sources carry a colour, so the bins hold about 22 and 17 stars each; thin, but all six survive. The V-I edges are -1.98, -0.34, -0.09, 0, 0.09, 0.21, 2.31 (BLG41) and -1.91, -0.42, -0.09, 0, 0.10, 0.21, 2.30 (BLG01). Equal-population quantile binning is invariant under any monotonic transformation of the colour, so a colour-scale error cannot move which star lands in which bin.

Only the *differential* part is recoverable. Refraction common to every star is an identical displacement of the whole field, which is exactly a translation and is absorbed silently by `tx`, `ty` at every epoch. What the fit measures is how much more the red stars are refracted than the blue ones; the mean refraction of the field is gone and cannot be retrieved.

Annual term. A quartic in the phase of the calendar year, $p = (\text{JD} - \text{year start})/(\text{year length}) - 0.5$, giving `[1, p, p2, p3, p4]` per axis — **10 parameters, fitted globally**, one curve shared by every star (`ParA` is $10 \times N_{\text{src}}$ but holds a single distinct column). It absorbs whatever repeats with the seasons and is common to the field: the annual cycle of observing geometry, temperature, and the airmass at which the field is reachable. It is **not** parallax, which is

per-source and switched off.

Pixel-phase term. CCD response varies within a pixel, so a measured centroid is pulled toward or away from the pixel centre according to where the light falls. Modelled as a quintic in the sub-pixel phase, $[\varphi, \varphi^2, \varphi^3, \varphi^4, \varphi^5]$ per axis — **10 parameters, again global** (ParPix is 10×1). It is recoverable only because the pipeline stored the per-epoch registration shifts; without them the phase is unknown and the fit refuses to run with this term on.

SysRem. Two components on the astrometric residuals, computed **once over the whole decade** and reused, not refitted per season. On the reference reduction this change cut the target's season-to-season scatter from 5.32 to 2.28 mas, because fitting seasons independently let each float on its own systematics. In v5 the correction is 18.4 / 22.5 mas rms and is fully dense — every epoch-source cell is finite — because these sources are bright and almost always detected. Point-wise rejection is applied so that a single bad cell is zeroed rather than the whole source discarded.

Weighting and outliers. Each source is weighted by its own residual scatter, recomputed every iteration, with moving-median outlier rejection over a 30-epoch window.

What is deliberately absent. Parallax is off (`Plx = false`): at bulge distances it is ≤ 0.12 mas, far below the noise floor. No microlensing or astrometric-deviation model is fitted — the solution is model-free, so no theoretical curve is imposed on the target. And no absolute astrometry enters the fit itself; the frame is relative, and Gaia is applied only afterwards as described in section 3.

Superseded — absolute proper motions. Every Gaia tie made before the v6 reduction removed only the **constant** part of the gauge. Relative astrometry leaves the proper motions free up to a constant *plus a linear gradient across the field*, and the residual gradient is a shear of about -0.036 mas/yr per pixel, found at nearly the same value in both fields. Removing the full six-parameter gauge cuts the scatter of our motions about Gaia from 4.34 / 3.57 to 0.98 / 0.66 mas/yr (BLG41) and 3.48 / 3.20 to 1.00 / 0.70 (BLG01), and moves the target's motion by a few tenths of a mas/yr. **All absolute proper motions in this document are therefore superseded by KMT260058_v6_report.pdf**, which gives -2.017 ± 0.335 / -7.163 ± 0.360 (BLG41) and -1.781 ± 0.256 / -7.136 ± 0.377 (BLG01) mas/yr. Relative motions, residual scatters and everything else here are unaffected.

5. Results

	BLG41 ΔX / ΔY	BLG01 ΔX / ΔY
calibration stars (126 / 96)	4.87 / 4.81	4.77 / 4.77
the target	25.61 / 22.27	28.45 / 23.15

mas. **4.8 mas is the best frame-star astrometry this project has produced** — better than the reference reduction's bright stars (6.41 / 6.88 and 6.48 / 7.01) and better than the same objects pinned to Gaia in v4 (6.13 / 5.20 and 5.30 / 5.42). Some of that advantage is self-fulfilling: a free fit adjusts four parameters per star and so absorbs part of each star's own systematics.

The target is **not** improved. The reference reduction measures it at 24.62 / 21.00 and 25.48 / 21.78, better on every axis in both fields. Restricting the analysis set costs 4 to 12% on the target while improving the calibration stars, because the per-epoch frame is now built from 126 stars instead of ~600 weighted sources.

Target proper motion, absolute, gauge removed:

	$\mu_\alpha \cos \delta$	μ_δ
BLG41	-1.80	-6.20
BLG01	-2.13	-6.51

mas/yr; the two fields differ by 0.33 / 0.31. Using `gaiaTie`'s own matrix instead gives $-1.978 / -6.757$ and $-1.906 / -6.880$, differing by 0.072 / 0.123. The honest statement is that the fields agree to **0.07 to 0.36 in RA and 0.12 to 0.29 in Dec**, the spread being the choice of tie method.

6. Comparison stars

Six per field, all freely fitted, all plotted in the same style as the target:

tag	I	distance from the target	decade residual RMS, BLG41 / BLG01
cal_d24	16.68	24 pix	4.07 / 4.14
cal_d30	16.63	30 pix	6.73 / 6.83
cal_d43	16.94	43 pix	4.48 / 4.60
out_d04	18.14	4.5 pix	28.61 / 32.37
out_d21	17.91	21 pix	17.36 / 17.54
out_d48	18.05	48 pix	19.12 / 18.59

Coordinates. Pixel positions are per field, since the two cut-outs are offset by about 1.1 pixels; sky positions are derived by inverting each field's own Gaia-to-pixel map, and the two fields agree on them to **2 to 9 mas**, which is an independent check on both maps.

star	I_OGLE	dist [pix]	BLG41 pixel	BLG01 pixel	RA (J2000)	Dec (J2000)
target	18.13	0.0	150.4, 150.4	151.6, 150.2	17:52:38.085	-31:47:36.21
cal_d24	16.68	24.5	161.3, 172.2	162.4, 172.1	17:52:37.746	-31:47:27.53
cal_d30	16.63	30.5	120.4, 154.6	121.4, 154.5	17:52:39.023	-31:47:34.60
cal_d43	16.94	42.9	110.9, 166.6	111.9, 166.5	17:52:39.318	-31:47:29.89
out_d04	18.14	4.5	146.4, 148.3	147.6, 148.0	17:52:38.210	-31:47:37.06
out_d21	17.91	21.3	155.3, 171.1	156.4, 170.9	17:52:37.935	-31:47:28.01
out_d48	18.05	48.5	198.6, 145.4	199.8, 145.0	17:52:36.585	-31:47:38.10

The target's recovered position, 17:52:38.085 -31:47:36.21, sits 0.11 arcsec from the OGLE EWS coordinates the pipeline was pointed at, which is the accuracy of the centroid plus the map, not a disagreement.

mas in X. The three `cal_` stars are members of the calibration set, chosen from those present in **both** fields so the same physical object carries the same name everywhere. The three `out_` stars sit at the target's own magnitude and are deliberately outside the set, to give something comparable to the target.

out_d04 is not a usable comparison object. It lies 4.5 pixels from the target under a seeing FWHM near 7 pixels, so it shares most of its light with it. Its residual RMS is 29 to 32 mas against 17 to 19 for the other two stars of the same brightness, in every season and not only during the event; and its 2026 season is displaced by 40 to 58 mas as the target brightens. Its CSV carries a warning in the header. `out_d21` and `out_d48` are the honest controls, and their errors bracket the target's own 25 to 27 mas — the check that the target is not anomalously noisy for what it is.

7. Files

file	content
report_v5_RMS.png	residual RMS against OGLE I, per axis and field, for this run
report_v5_motion_<field>_target.png	the target, sidereal-month bins, 2x2
report_v5_motion_<field>_cal_d{24,30,43}.png	the three calibration comparison stars
report_v5_motion_<field>_out_d{04,21,48}.png	the three I ~ 18 comparison stars
source_motion_v5_<field>.csv	the target, per epoch
source_motion_v5_<field>_<tag>.csv	each comparison star, per epoch

CSV columns are JD, X_mas, Y_mas, errX_decade, errY_decade, errX_season, errY_season, season. Positions are relative to that star's own mean, with **proper motion not removed**. *err_decade is that star's own residual RMS over the whole run and err_season the same within each season*; note this differs from the reference reduction's CSV, where the decade column held a field-star population estimate — v5 keeps no field stars at I ~ 18, so that curve cannot be formed here.

In every motion figure the columns are the two axes, **X left and Y right**; the top row keeps the proper motion and the bottom row removes it. The legend quotes the motion along the pixel axis and the heading quotes it on the sky with the gauge removed. **Pixel X runs opposite to RA**, so those two carry opposite signs in X by construction.

Figures

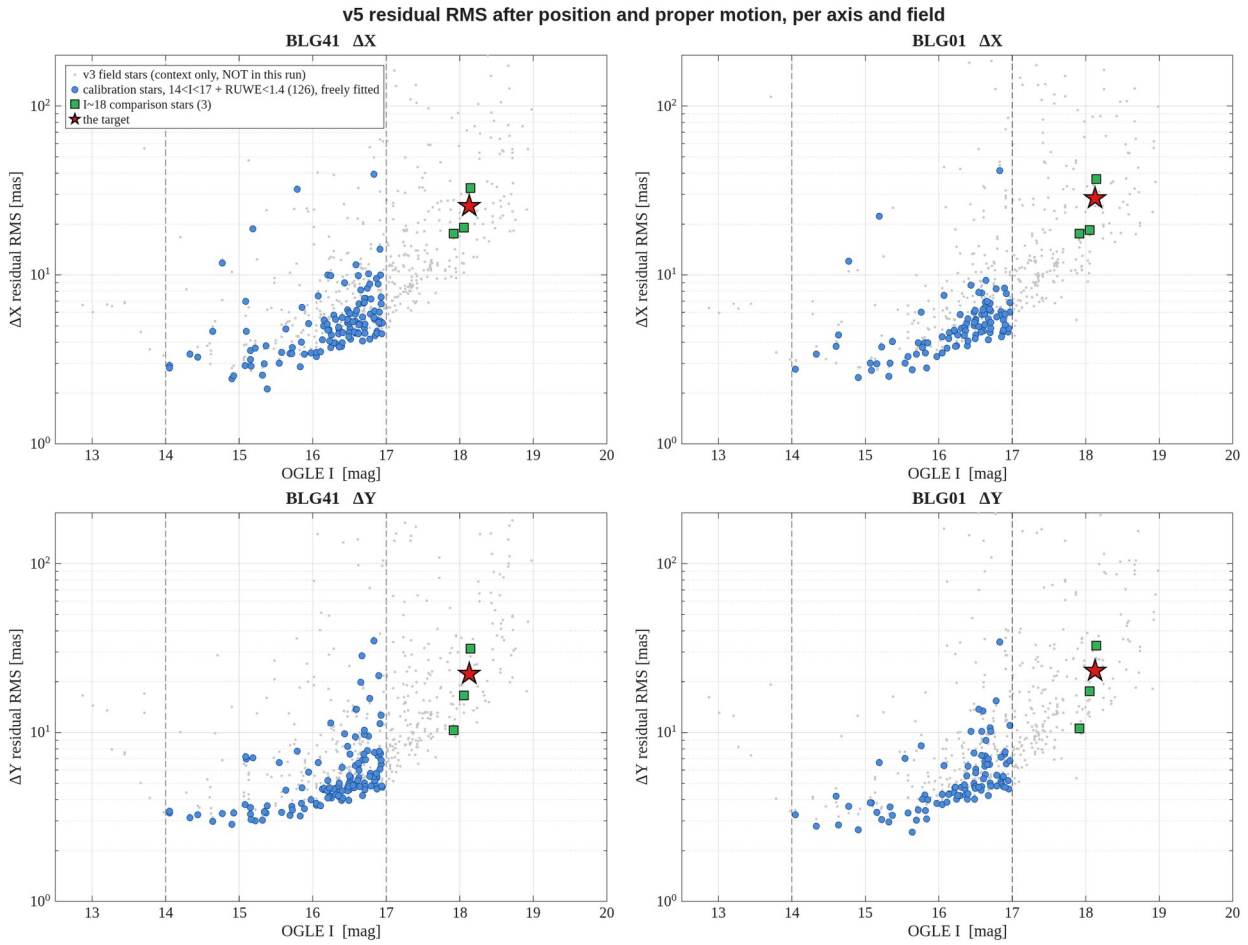


Figure 1. Residual RMS against the OGLE catalogue magnitude, per axis and per field, for the v5 run, after position and proper motion have been removed. Blue: the 126 / 96 calibration stars, 14 < I < 17 with RUWE < 1.4, **freely fitted**. Green: the three I ~ 18 comparison stars. Red: the target. Grey: field stars from the reference reduction, shown for context only — **they are not part of this run**. Dashed lines mark the 14 and 17 mag limits.

report_v5_RMS.png

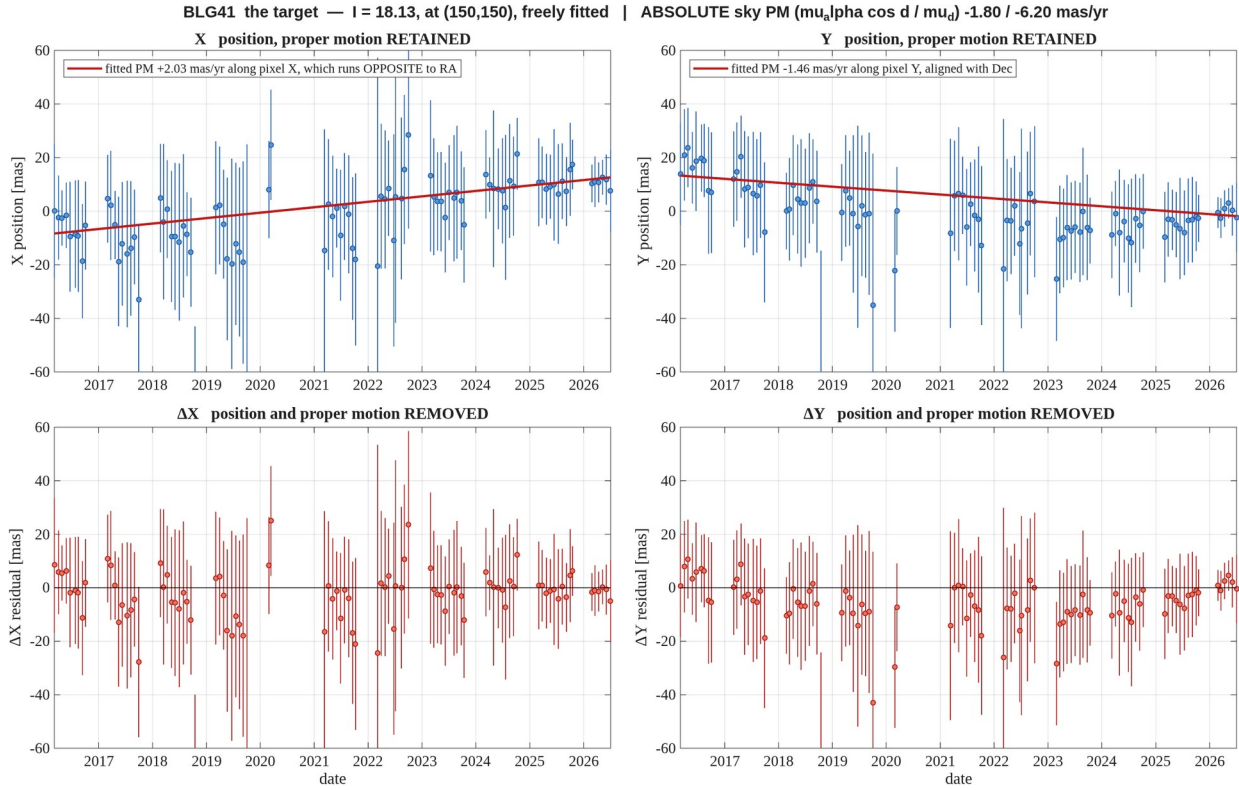


Figure 2. **BLG41** — the target, $l = 18.13$. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG41_target.png

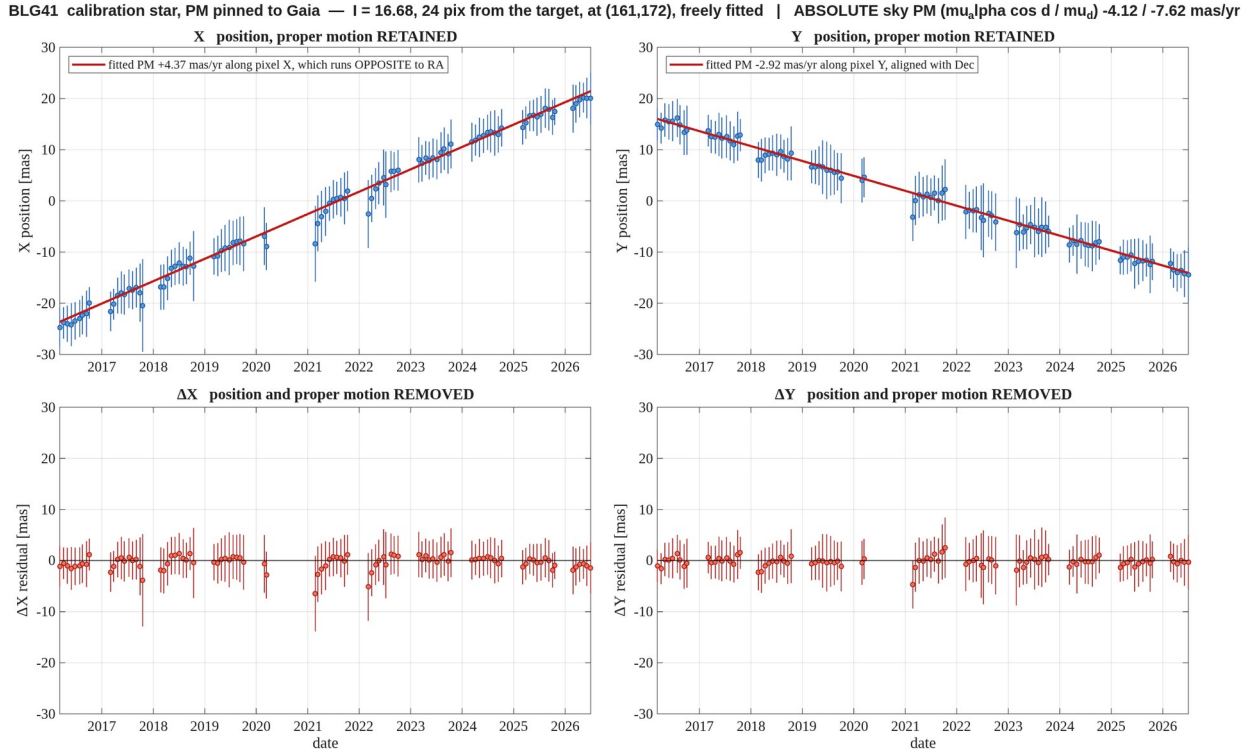


Figure 3. **BLG41** — calibration star, $l = 16.68$, 24 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG41_cal_d24.png

BLG41 calibration star, PM pinned to Gaia — $l = 16.63$, 30 pix from the target, at (120,155), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \lvert \cos d / \mu_\delta$) $+2.20 / +1.95$ mas/yr

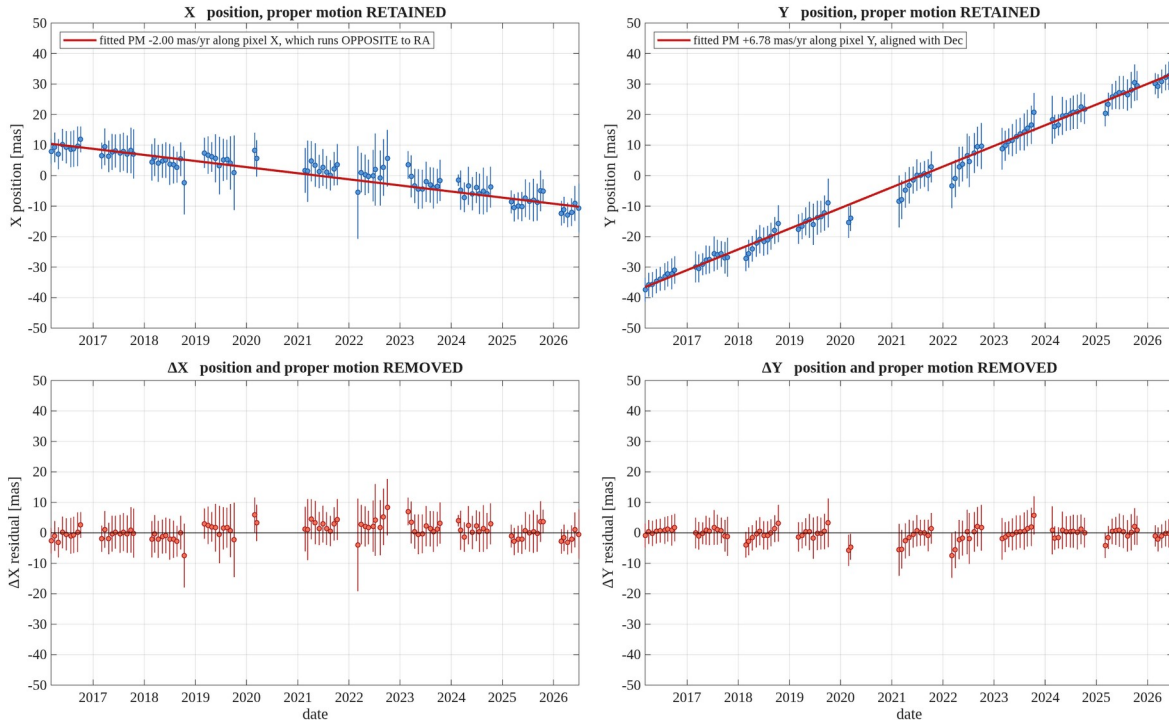


Figure 4. **BLG41** — calibration star, $l = 16.63$, 30 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG41_cal_d30.png

BLG41 calibration star, PM pinned to Gaia — $l = 16.94$, 43 pix from the target, at (111,167), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \lvert \cos d / \mu_\delta$) $-0.28 / -5.96$ mas/yr

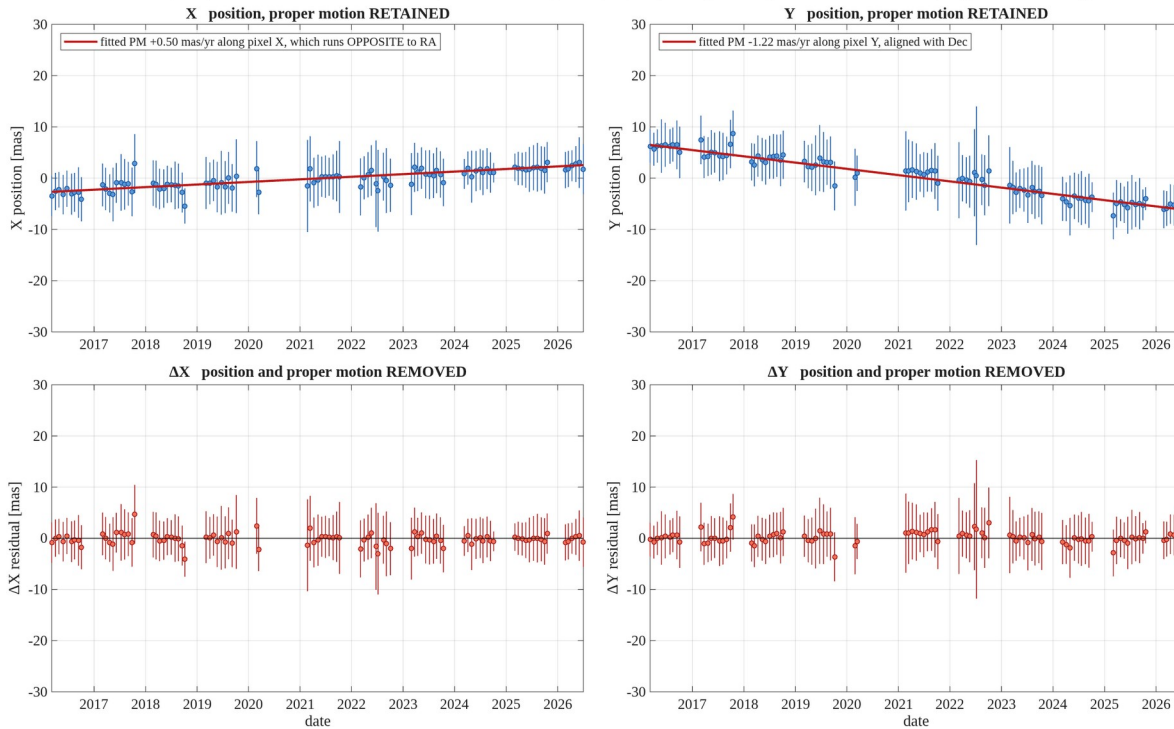


Figure 5. **BLG41** — calibration star, $l = 16.94$, 43 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG41_cal_d43.png

BLG41 comparison star outside the set, PM fitted — $l = 18.14$, 5 pix from the target, at (146,148), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) $-0.17 / -4.55$ mas/y

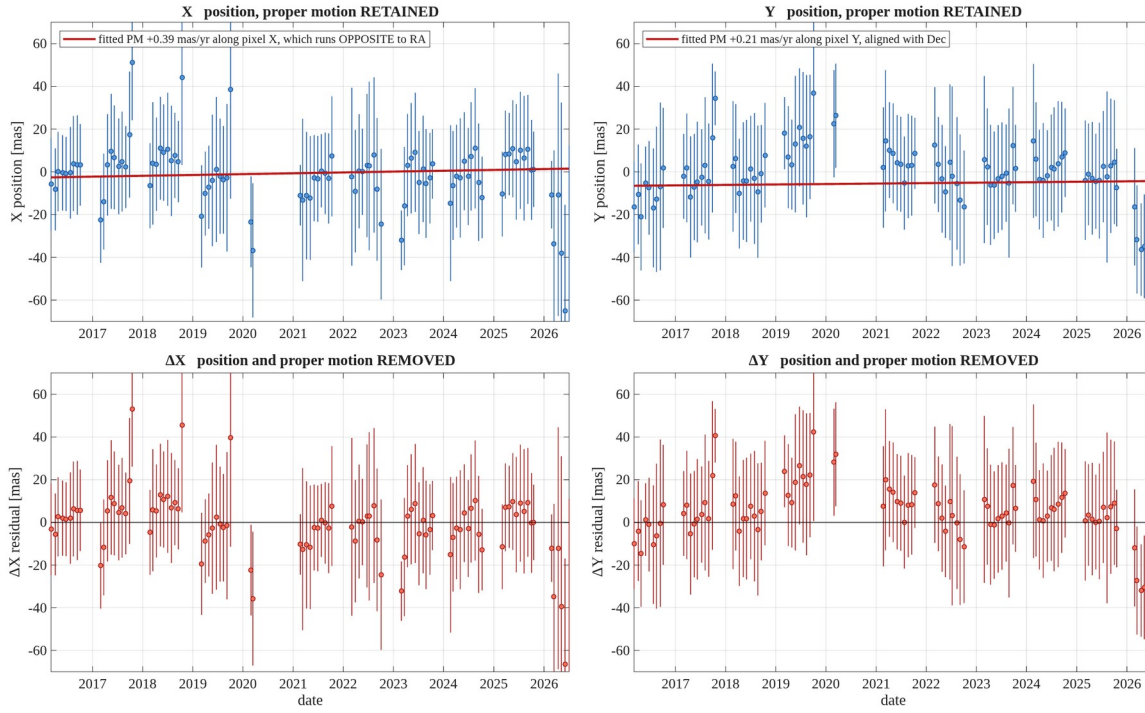


Figure 6. **BLG41** — comparison star $l = 18.14$, 4.5 pix away — **blended with the target**, see section 6. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X. report_v5_motion_BLG41_out_d04.png

LG41 comparison star outside the set, PM fitted — $l = 17.91$, 21 pix from the target, at (155,171), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) $-7.71 / -9.55$ mas/y

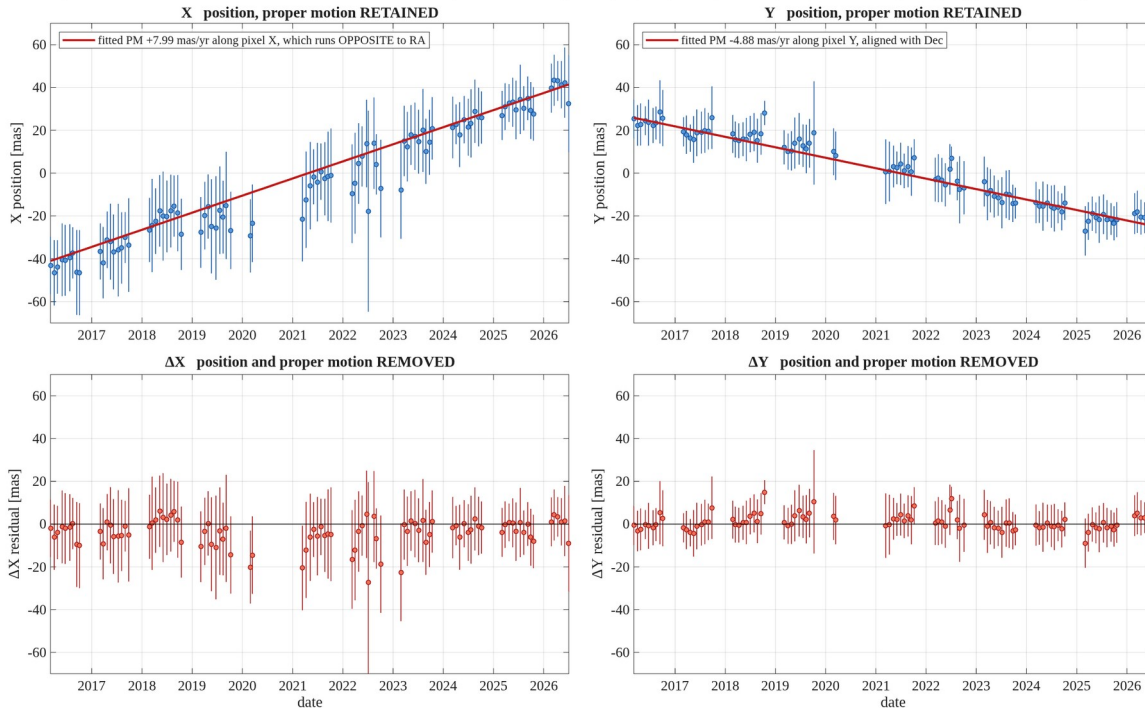


Figure 7. **BLG41** — comparison star $l = 17.91$, 21 pix away. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X. report_v5_motion_BLG41_out_d21.png

BLG41 comparison star outside the set, PM fitted — $l = 18.05$, 48 pix from the target, at (199,145), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) +3.08 / -7.56 mas/

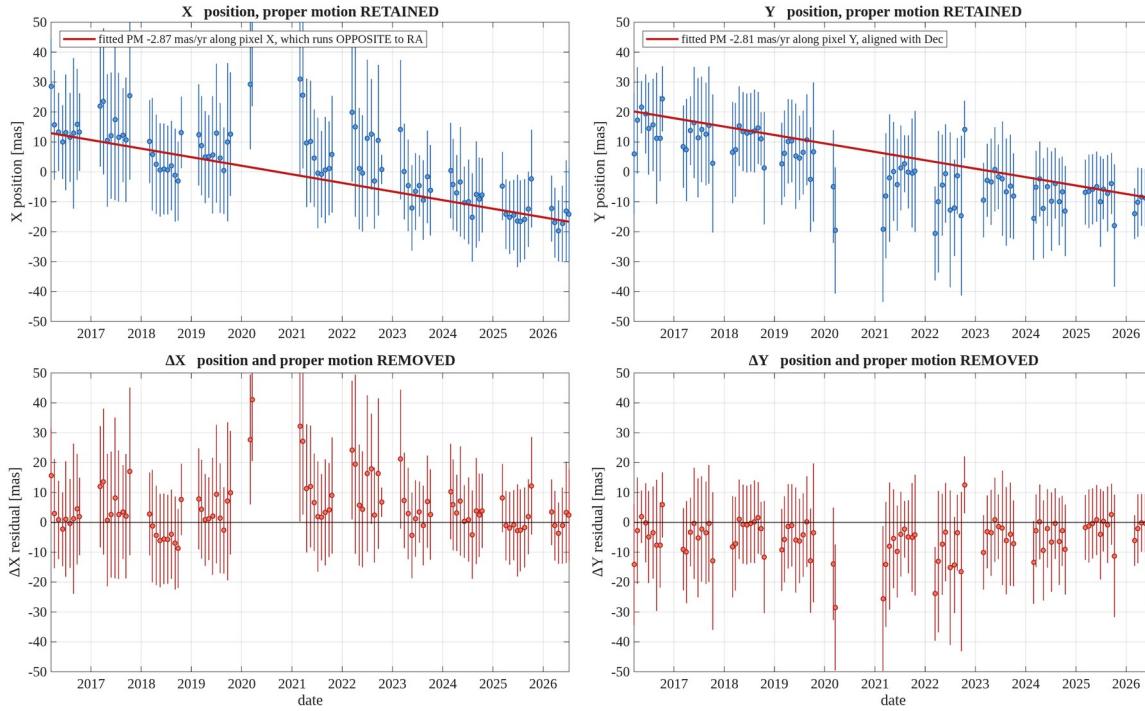


Figure 8. **BLG41** — comparison star $l = 18.05$, 48 pix away. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG41_out_d48.png

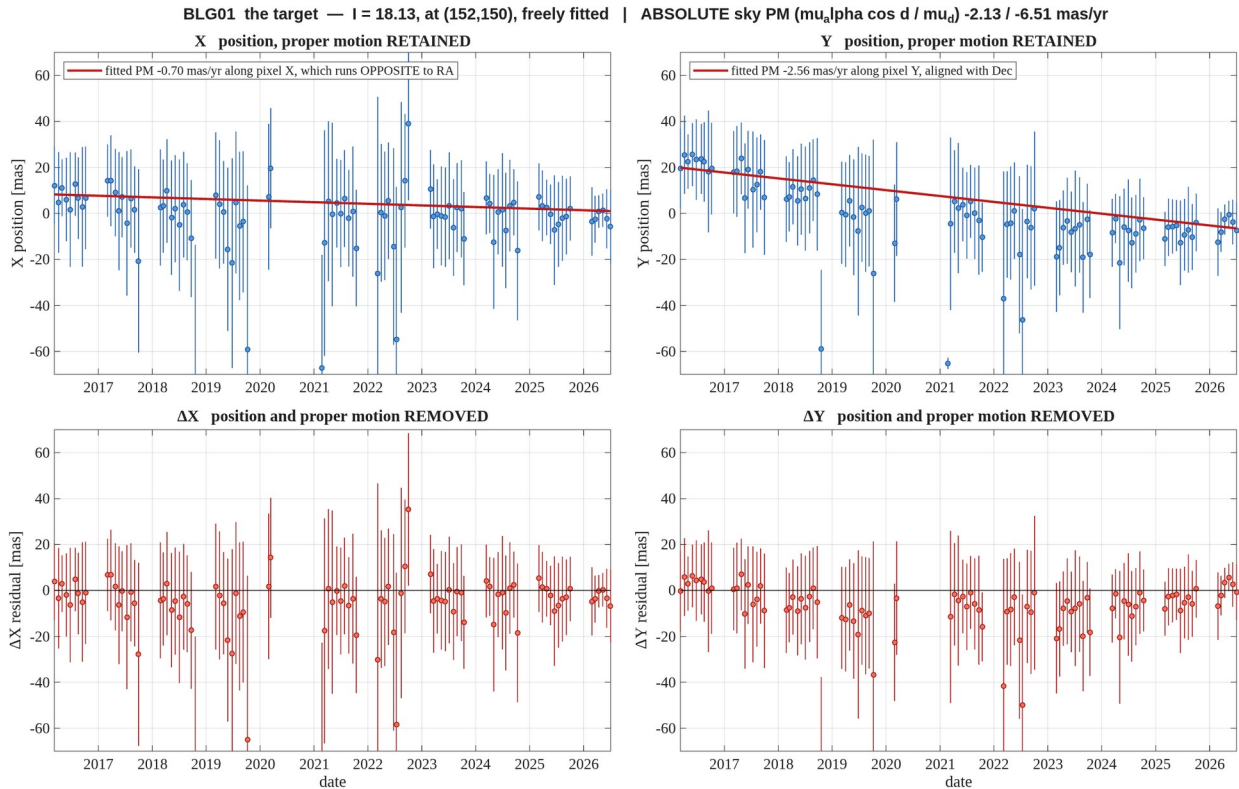


Figure 9. **BLG01** — the target, $l = 18.13$. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_target.png

BLG01 calibration star, PM pinned to Gaia — $l = 16.68$, 24 pix from the target, at (162,172), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) -4.59 / -7.73 mas/yr

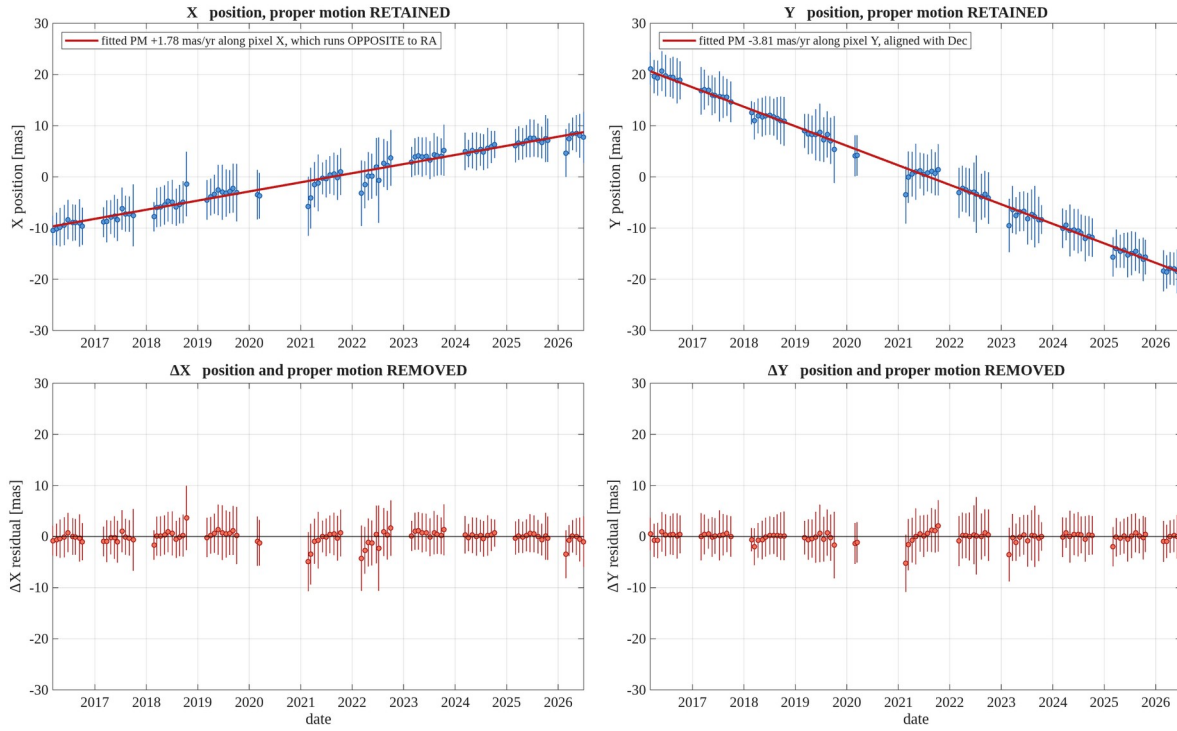


Figure 10. **BLG01** — calibration star, $l = 16.68$, 24 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_cal_d24.png

BLG01 calibration star, PM pinned to Gaia — $l = 16.63$, 30 pix from the target, at (121,154), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) +1.58 / +1.97 mas/yr

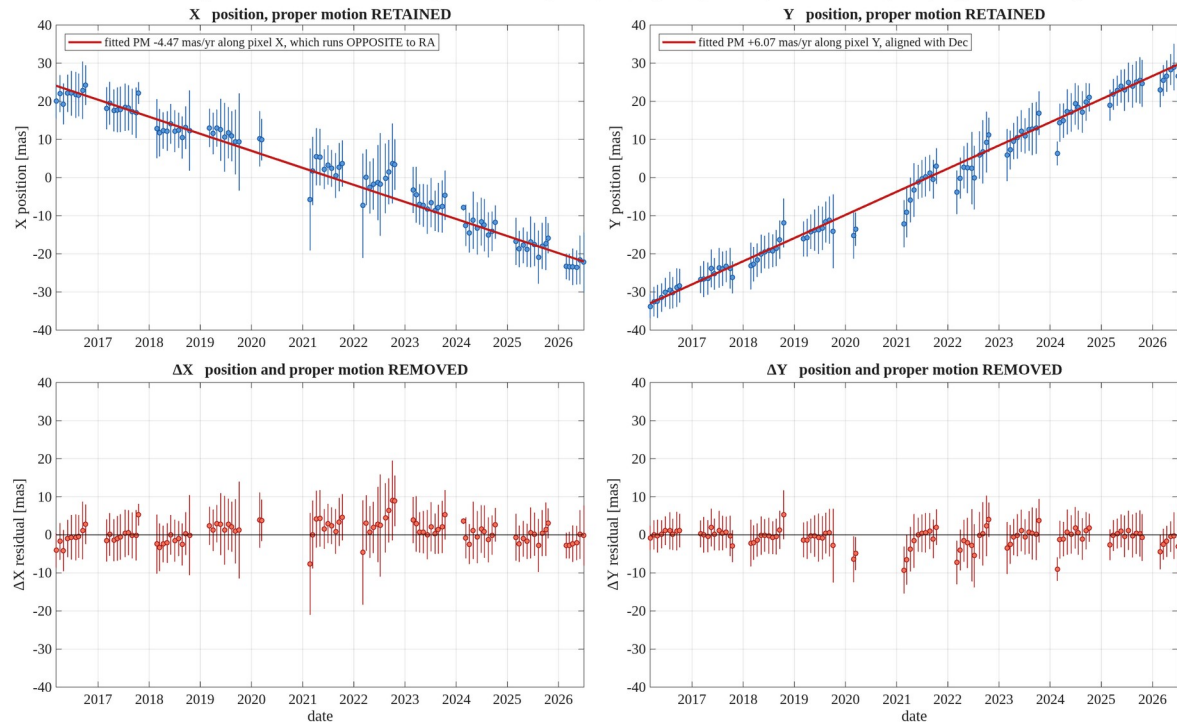


Figure 11. **BLG01** — calibration star, $l = 16.63$, 30 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_cal_d30.png

BLG01 calibration star, PM pinned to Gaia — $l = 16.94$, 43 pix from the target, at (112,166), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) -1.00 / -5.99 mas/yr

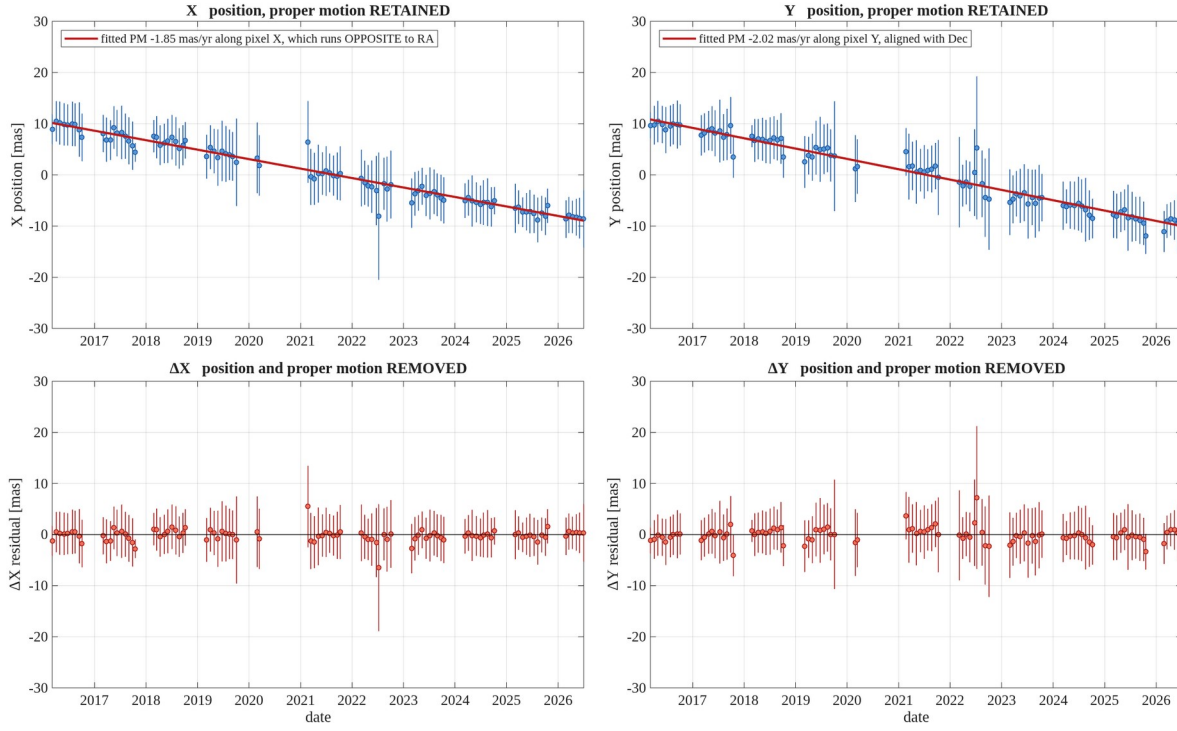


Figure 12. **BLG01** — calibration star, $l = 16.94$, 43 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_cal_d43.png

BLG01 comparison star outside the set, PM fitted — $l = 18.14$, 5 pix from the target, at (148,148), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) -0.71 / -5.42 mas/yr

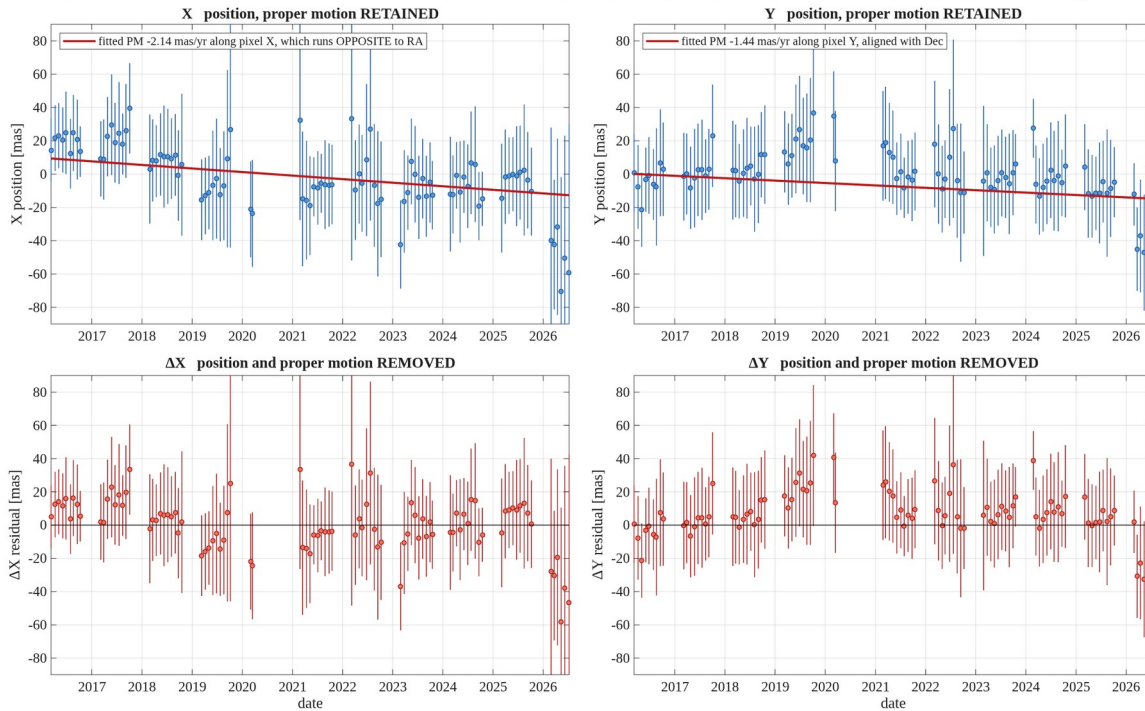


Figure 13. **BLG01** — comparison star $l = 18.14$, 4.5 pix away — **blended with the target**, see section 6. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_out_d04.png

LG01 comparison star outside the set, PM fitted — $l = 17.91, 21$ pix from the target, at (156,171), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) $-8.49 / -9.74$ mas/

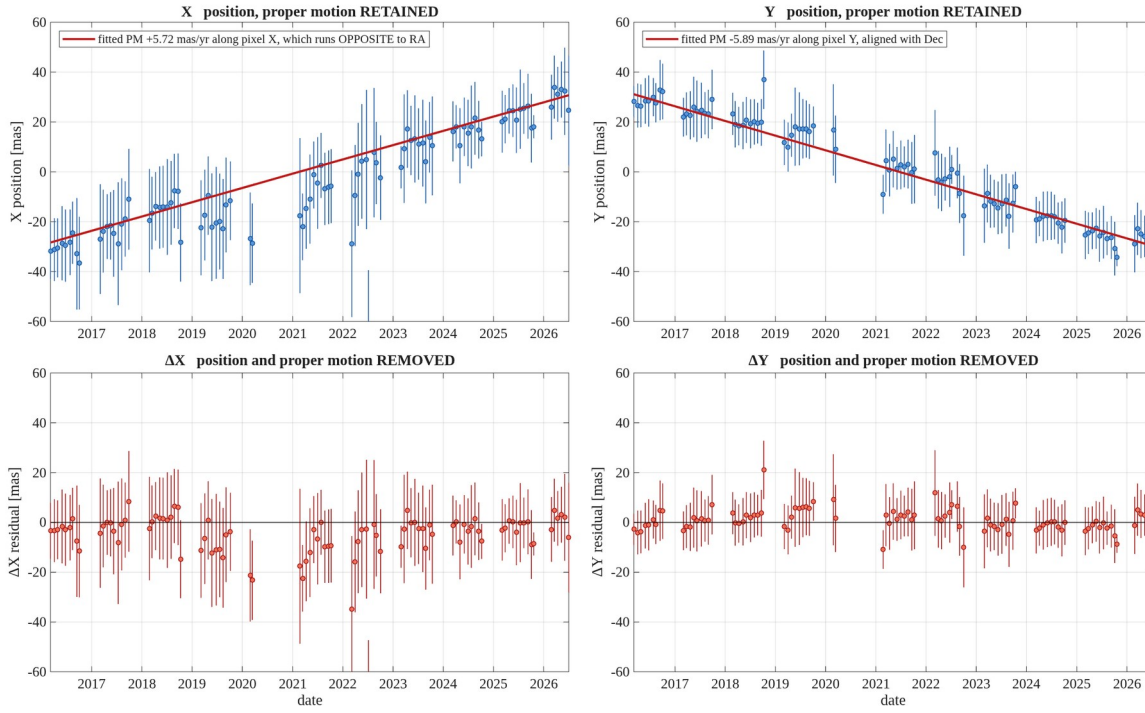


Figure 14. **BLG01** — comparison star $l = 17.91, 21$ pix away. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_out_d21.png

LG01 comparison star outside the set, PM fitted — $l = 18.05, 49$ pix from the target, at (200,145), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) $+2.83 / -7.32$ mas/

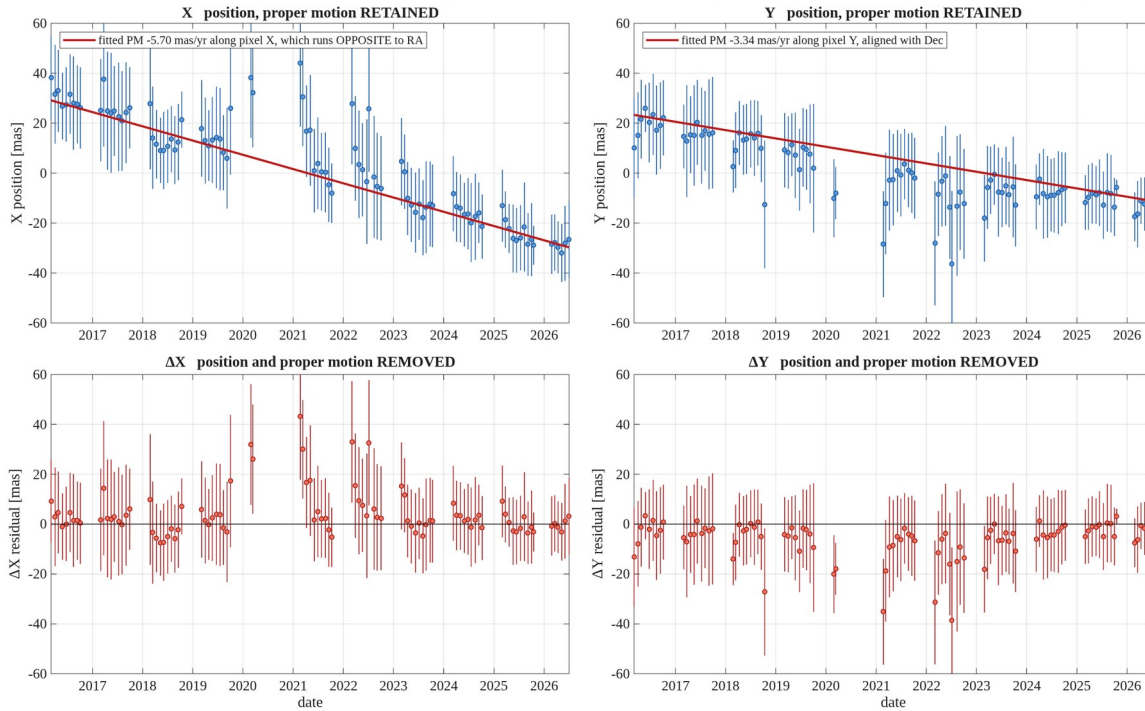


Figure 15. **BLG01** — comparison star $l = 18.05, 48$ pix away. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_out_d48.png